

Week 4 Activity Log

Task 2 – To-Do Application & Advanced JavaScript Practice

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Internship Program: CoderAxo Web Development Internship

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Week Duration: July 6 – July 12, 2026

Week Overview

With the portfolio behind me, this week was built entirely around Task 2 — a fully interactive To-Do application. The scope was bigger than the portfolio's JavaScript: task creation and deletion, a completed-state toggle, persistent storage, and a dark/light theme switch, all wrapped in a responsive UI.

Objectives for the Week

- Build a fully functional CRUD-style app entirely with vanilla JavaScript
- Learn to persist application data on the client using localStorage
- Implement a theme toggle using CSS variables rather than duplicate stylesheets

Daily Activity Breakdown

Mon, Jul 6

Reviewed Task 2 requirements in detail: Add Task, Delete Task, Mark Completed, Save to localStorage, Dark/Light Mode, Responsive UI.

Tue, Jul 7

Built the base HTML/CSS layout for the app and implemented Add Task using an input field and a submit handler.

Wed, Jul 8

Implemented Delete Task and Mark as Completed using event delegation and classList toggling for the completed style.

Thu, Jul 9

Wired up localStorage — saving the task array as JSON on every change and loading/parsing it back in on page load.

Fri, Jul 10

Built the dark/light mode toggle using CSS custom properties and a class switch on the body element.

Sat, Jul 11

Made the layout responsive for smaller screens, tested across breakpoints, and refactored the JS into smaller, named functions.

Sun, Jul 12

Final testing and bug fixes (task list occasionally not refreshing correctly), then submitted Task 2 with the Week 4 log.

Detailed Summary

The core challenge in this task wasn't any single feature on its own — it was keeping the on-screen task list, the underlying JavaScript array, and the localStorage copy all in sync every time something changed. My first attempt tried to update the DOM directly for each add/delete/complete action, which quickly got messy and occasionally left the list out of sync with localStorage after a page refresh. I ended up rewriting the render logic so that any change to the task array triggers one function that clears and fully re-renders the list from that array, then saves it to localStorage right after. That one change made the whole app far more predictable.

Tools & Technologies Used

- VS Code
- Vanilla JavaScript (ES6)
- Browser localStorage / Application tab in DevTools
- CSS custom properties (variables)

Skills & Concepts Learned

- Persisting client-side data with localStorage, including JSON.stringify/JSON.parse
- Using CSS variables to build a theme toggle without duplicating stylesheets
- Event delegation for handling clicks on dynamically created list items
- Refactoring a growing script into smaller, single-purpose functions

Challenges Faced & How I Resolved Them

The sync issue between the DOM, the in-memory array, and localStorage was the main obstacle, described above. There was also a smaller bug where deleting a task sometimes deleted the wrong one — this was caused by using the array index instead of a unique task ID during deletion, which broke once tasks were reordered. Switching to a generated unique ID per task fixed it permanently.

Reflection

This task taught me the value of having one clear source of truth (the array) and always rendering the UI from it, rather than trying to patch the DOM piecemeal. It's a pattern I now default to even outside of vanilla JS.

Goals for Next Week

- Move into Node.js and start understanding how JavaScript works outside the browser
- Get comfortable with npm and package.json before Express is introduced
- Carry the 'single source of truth, re-render from state' habit forward into backend-connected apps